

OMER BIN HABIB

Software Engineer · AI features · LLM Integration · Python · TypeScript · Node.js · PostgreSQL & pgvector

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SUMMARY

Software engineer with six years of professional experience, backend-focused, built mainly on Node.js, TypeScript and PostgreSQL, with Python (FastAPI) sidecar. Built LLM and NLP features into backend systems, including tool-use workflows with grounding constraints, schema-validated structured output, pgvector hybrid retrieval, and a FastAPI service for PII redaction, outlier scoring, and sentiment gating. Treats model output as untrusted and designs systems to verify it before it affects application behaviour.

TECHNICAL SKILLS

Backend: Node.js, TypeScript, NestJS, Express, REST API design, JWT authentication, input validation, Swagger / OpenAPI, queue-based processing (BullMQ, Redis), WebSockets (Socket.io), scheduled and background jobs, idempotency, retry and backoff, transactional consistency, bounded concurrency

Databases: PostgreSQL and MySQL — data modelling, normalisation, indexing, views, migrations, stored procedures, window functions, multi-tenant schemas with row-level security

Other server-side languages: PHP (Laravel, Lumen), Python (FastAPI)

Cloud & DevOps: AWS — Lambda, RDS, S3, EC2; Docker, GitHub Actions, nginx, Render, Vercel, Supabase, Sentry, Git

Design & practice: Object-oriented design and dependency injection through NestJS, design patterns, code review, automated testing (Jest / Vitest, Mocha, Playwright end-to-end), ESLint

Frontend: React, Next.js (App Router), Redux, TypeScript, Tailwind

AI & LLM: Claude, OpenAI and Gemini APIs; tiered model routing — a cheaper model first, a stronger one only for ambiguous or high-stakes cases; schema-validated structured output (Zod); closed-set validation of model-supplied values; PII redaction before external calls; evaluation-gated model changes; prompt caching; pgvector with hybrid semantic and keyword retrieval; LLM tool-use loops with a grounding rule; LangGraph multi-agent workflows with Langfuse tracing; NLP with spaCy, scikit-learn and sentence-transformers

AI-assisted development: Claude Code as the main development method, under a written RULES.md the sessions are held to (reproduce before fixing, schema truth from migrations, no unverified numbers); project-scoped skills, sub-agents, slash commands, a PreToolUse hook and a tree-sitter code graph

Payments: Stripe Connect tiered billing

EXPERIENCE

Independent Consultant · Remote · Karachi, Pakistan

Jan 2026 – Present

Backend engineer for UK SaaS clients, working asynchronously with distributed clients

- **Spot — UK B2B workplace wellbeing SaaS.** TypeScript and NestJS backend. Stripe Connect tiered billing, timezone-aware processing, and data handling built for GDPR from the start with hard delete, anonymisation and export flows. Alongside it I built a Python FastAPI NLP service: PII redaction with spaCy named-entity recognition and language detection, outlier scoring with an Isolation Forest, and sentiment gating with sentence embeddings. 357 automated tests and 18 database migrations maintained across the life of the project.
- **AgencyView — reporting platform for a UK SaaS founder.** PostgreSQL-driven engine producing thirteen report types with output filtered by user role. Ingestion runs as a scheduled pipeline with bounded concurrency, retry policies and structured logging, so failures are diagnosable without manual inspection.
- **Remote and asynchronous by default — requirements, trade-offs and release scope agreed in writing with clients in other timezones.**

Software Engineer · Blocship · Karachi, Pakistan

May 2023 – Dec 2025

Backend development across six products, in a cross-functional team with frontend, product, design and QA

- **QR-code points and loyalty system for a vehicle-parts distributor.** NestJS backend handling point issuance, redemption and transfer between users — balance changes written transactionally so a failed transfer could not leave value in two places at once. I built the points reporting the distributor used to pay its buyers back according to the volume they had bought.
- **MaidMe — home services marketplace for providers in Norway.** The Laravel backend was already live and had no support for socket connections, so rather than rewrite it I added a dedicated NestJS socket server alongside it — bringing real-time booking management and in-app support chat to a running product.
- **Groops — social platform for organisations.** NestJS alongside a Lumen service, covering group membership and engagement features — built on Clean Architecture: 16 NestJS modules, 73 migrations, 81 entities and 472 test suites.
- **Hubfiit and Apna Khata** — fitness and nutrition tracking; and an app for Pakistan's produce markets, where farmers sold fruit and vegetables to market traders — I built the transaction tracking between them. Laravel backends.
- **Across all six:** REST API design documented with Swagger, JWT authentication and input validation, MySQL and PostgreSQL data modelling and migrations, code review, and production support — reproducing reported faults, tracing them through logs and shipping fixes under live conditions.

Software Engineer · Sybrid (Pvt) Ltd · Karachi, Pakistan

Apr 2021 – May 2023

Enterprise web applications for Japanese and Pakistani clients

- **FITE — IT engineer hiring portal for Japanese companies.** Laravel, fully bilingual in English and Japanese, with role-based access across four user groups and security aligned to OWASP guidance.
- **MySQL stored procedures using window functions** to persist candidate profiles and calculate a profile score server-side, keeping scoring logic in one place rather than spread across request handlers.
- **Containerised a scheduled email service with Docker** — a reminder job for candidates who had begun but not completed an application, running behind nginx so the environment was reproducible rather than dependent on host configuration.

Associate Developer · TAFSOL Technologies · Karachi, Pakistan

Sep 2020 – Apr 2021

- **Digital sports agency platform** — Laravel, with payment gateway integration, an e-store, a training session video channel and an admin panel. Employee of the Month, October 2020.

SELECTED PROJECTS

ALLDost · Activity and nutrition tracking web application

Personal product

- **Next.js and React on a Supabase PostgreSQL backend** with row-level security, and an automated test suite maintained alongside it, including Playwright end-to-end tests.
- **Natural-language parsing of meals and workouts** into structured records using the Gemini API, with output validated against Zod schemas and values the model supplies checked against a fixed set rather than trusted.
- **Evaluation and review layer:** every AI-generated food entry is held as unverified until a reviewer approves or corrects it through a purpose-built review screen with its own access rules, covered by end-to-end tests.

Multi-agent build harness · LangGraph and Langfuse

Personal project

- **Orchestrator, specialist and critic agents that build software**, traced in Langfuse. Used it to build resume-tailor (public on GitHub), a resume generator that only includes claims backed by a verified profile.

EDUCATION

Master's in Business Analytics (MSBA) · Karachi School of Business & Leadership

2025 – 2026

Final semester (*In Progress*)

BSc Computer Science · SZABIST · Karachi

Graduated 2020

CERTIFICATIONS

Introduction to Cloud Computing — IBM (2025) · Generative AI with Large Language Models — DeepLearning.AI with AWS · Microsoft Power BI · Building RESTful APIs in Laravel — LinkedIn Learning (2022)